

Notes: Boehm & Oberfield (QJE, 2020)

Misallocation in the Market for Inputs: Enforcement and the Organization of Production

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1 Big picture

- **Question.** How does weak contract enforcement distort firms' sourcing decisions, vertical organization, and aggregate productivity?
- **Setting.** Indian formal manufacturing, where courts are notoriously slow and backlogged. The paper uses cross-state variation in High Court congestion and detailed plant-level input-output data from the Annual Survey of Industries (ASI).
- **Empirical core.** In states with slower courts, plants in industries that rely more on relationship-specific inputs use fewer intermediate inputs overall, tilt their materials baskets toward homogeneous inputs, and perform more production stages in-house.
- **Conceptual core.** The paper models firms as choosing among multiple *recipes*—alternative technology/organization bundles with different input sets. Weak enforcement raises the effective cost of relationship-specific inputs, so firms respond both on the **intensive margin** (buy less of a given input) and on the **extensive margin** (switch suppliers or switch recipes).
- **Key identification idea.** Among firms using the same recipe, average cost shares are invariant to factor prices but depend on distortions. This lets the authors identify enforcement wedges from cross-state differences in first moments of cost shares, while still allowing large within-industry heterogeneity in technologies.
- **Main quantitative result.** Reducing court congestion to the level of the best state raises manufacturing productivity by about **4% on average** and by **high-single digits** in the most congested states. Halving the estimated wedges on relationship-specific inputs raises productivity by **7.1% on average**, with much larger gains in the most distorted states.
- **Why this matters.** The paper connects legal institutions to production networks and misallocation. Weak enforcement does not merely shrink firm-to-firm trade; it changes the organization of production itself, and those micro distortions aggregate into meaningful losses in productivity.

2 Data and measurement (key objects)

2.1 Plant-level production data

- **Source.** Annual Survey of Industries (ASI), 2000/01–2012/13.
- **Coverage.** The ASI covers all establishments with more than 100 employees and each year samples one-fifth of establishments with more than 20 employees (or more than 10 if they use power).
- **Key advantage.** The ASI records detailed product-level intermediate inputs and outputs at roughly the five-digit level, with about 5,200 product codes.
- **Main reduced-form sample.** Single-product plants, because for multiproduct plants the data do not say which input is used for which output.
- **Stylized fact from the raw data.** Even within very narrow product categories, plants use very different input bundles. That heterogeneity is a feature the model must accommodate rather than dismiss as pure noise.

2.2 Relationship-specific versus homogeneous inputs

The paper follows Rauch (1999) and classifies intermediate inputs into:

- **Homogeneous inputs:** traded on organized exchanges or with reference prices.
- **Relationship-specific inputs:** the residual category, where customization and holdup problems are more likely.

At the industry level, reliance on relationship-specific inputs is measured as the national fraction of intermediate-input expenditure devoted to relationship-specific inputs:

$$\text{RelSpec}_i = \frac{\sum_{j \in i} X_j^R}{\sum_{j \in i} (X_j^R + X_j^H)}. \quad (1)$$

- **Interpretation.** Industries with high RelSpec_i are the ones most exposed to court quality because they more often need enforceable supplier contracts.
- **Plant-level composition object.** A plant's tilt toward contract-sensitive inputs is

$$\frac{X_j^R}{X_j^R + X_j^H}.$$

2.3 Court congestion

- **Source.** Daksh microdata on pending civil cases in Indian High Courts.
- **Main measure.** For each state, the authors use the **average age of pending civil cases** in the High Court at the end of calendar year 2016.
- **Why High Courts?** Although contract cases may formally start in district courts, litigants often bypass them or appeal quickly, and High Courts effectively determine whether business contracts can be enforced in a timely way.
- **Cross-state variation.** Average age ranges from less than one year in Goa and Sikkim to about four and a half years in Uttar Pradesh and West Bengal; the cross-state average is about two and a half years.
- **Historical instrument.** The age of the High Court: older courts, especially those created under British rule, have had longer to accumulate backlogs; newer courts created after state splits begin with a clean slate.

2.4 Vertical span of production

To capture how many consecutive production stages a plant performs internally, the paper defines

$$\text{verticalSpan}_j = \sum_{\hat{\omega} \in \Omega} \frac{X_{j\hat{\omega}}}{\sum_{\tilde{\omega} \in \Omega} X_{j\tilde{\omega}}} \text{verticalDistance}_{\omega\hat{\omega}}, \quad (2)$$

where $X_{j\hat{\omega}}$ is plant j 's expenditure on input $\hat{\omega}$, ω is the plant's output, and $\text{verticalDistance}_{\omega\hat{\omega}}$ is the typical number of steps between input $\hat{\omega}$ and output ω in the plant-level input-output network.

- **Interpretation.** A higher vertical span means the plant buys more upstream inputs and therefore appears to perform more stages in-house.
- **Economic intuition.** If courts are bad, outsourcing a customized stage becomes risky, so plants may internalize more stages of production.

3 Models and empirical specifications (all main equations + interpretation)

3.1 Reduced-form empirical specifications

The main facts are summarized by regressions of the form

$$\text{MatShare}_{jisd} = \beta_1(\text{CourtCong}_d \times \text{RelSpec}_i) + \alpha_i + \delta_{\text{district}} + \Gamma' Z_{id} + \varepsilon_{jisd}, \quad (3)$$

$$\frac{X_j^R}{X_j^R + X_j^H} = \beta_2 \text{CourtCong}_d + \alpha_i + \delta_{\text{district}} + \Gamma' Z_d + \varepsilon_j, \quad (4)$$

$$\text{verticalSpan}_{jisd} = \beta_3(\text{CourtCong}_d \times \text{RelSpec}_i) + \alpha_i + \delta_{\text{district}} + \Gamma' Z_{id} + \varepsilon_{jisd}. \quad (5)$$

- **Dependent variables.** Materials share of cost, share of relationship-specific inputs in the materials basket, and vertical span.
- **Regressor of interest.** Court congestion, either directly or interacted with industry reliance on relationship-specific inputs.
- **Expected signs.** $\beta_1 < 0$, $\beta_2 < 0$, $\beta_3 > 0$.
- **Controls.** Five-digit industry fixed effects, district fixed effects, log district GDP per capita, and interactions of relationship specificity with state characteristics such as trust, corruption, caste fragmentation, and language fragmentation.
- **Interpretation.** Slow courts matter most precisely where supplier holdup is likely to matter most.

3.2 Demand, varieties, and recipes

Final demand is nested CES:

$$U = \left(\sum_{\omega \in \Omega} v_{\omega}^{1/\eta} U_{\omega}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \quad U_{\omega} = \left(\int_0^{J_{\omega}} u_{\omega j}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}. \quad (6)$$

- η is the elasticity of substitution across industries.
- ε is the elasticity of substitution across varieties within an industry.
- J_{ω} is the mass of firms in industry ω .

Each industry ω has a set of recipes $\mathcal{R}_{\omega}$. Recipe $\rho \in \mathcal{R}_{\omega}$ is a production function $G_{\omega\rho}(\cdot)$ using labor and a bundle of intermediate inputs $\hat{\rho}$. If firm j uses recipe ρ with labor l and supplier-specific inputs $x_{s_1}, \dots, x_{s_n}$, output is

$$y_j = G_{\omega\rho}(b_l l, b_{\hat{\omega}_1} z_{s_1} x_{s_1}, \dots, b_{\hat{\omega}_n} z_{s_n} x_{s_n}). \quad (7)$$

- **Assumption 1.** $G_{\omega\rho}$ has constant returns to scale and all inputs are complements.
- **Interpretation.** Recipes capture distinct technology/organization choices, such as whether a plant outsources a stage or performs it internally.

3.3 Contracting wedges

For each potential supplier of a relationship-specific input, the buyer draws an enforcement wedge $t_x \in [1, \infty)$ from distribution $T(t_x)$. If supplier s charges price p_s , then the buyer's effective cost is $t_x p_s$: the supplier receives p_s and the remaining $(t_x - 1)p_s$ is wasted on extra labor needed to fix imperfect customization.

For technique ϕ , effective input costs are

$$\lambda_{\hat{\omega}}(\phi) \equiv \min_{s \in S_{\hat{\omega}}(\phi)} \frac{t_{xs}(\phi)p_s}{b_{\hat{\omega}}(\phi)z_s(\phi)}, \quad \lambda_l(\phi) = \frac{t_l w}{b_l(\phi)}. \quad (8)$$

Given the dual unit-cost function $C_{\omega\rho}(\cdot)$, the firm's unit cost is

$$\min_{\rho \in \mathcal{R}_{\omega}} \min_{\phi \in \Theta_{j\rho}} C_{\omega\rho}(\lambda_l(\phi), \{\lambda_{\hat{\omega}}(\phi)\}_{\hat{\omega} \in \hat{\rho}}). \quad (9)$$

- **Economic meaning.** Bad courts act like an input-specific shadow tax on customized sourcing.
- **Why wedges are random.** Some pairs can enforce informally through kinship, repeated relationships, or social ties, so the relevant wedge varies across matches.
- **Two margins of response.** A firm can buy less of a given relationship-specific input, switch to another supplier, or abandon the recipe that uses that input.

3.4 Technology draws and recipe intensities

The key structural assumptions are:

- suppliers are drawn from the population of producers of each input;
- match-specific supplier productivity draws follow a power-law/Poisson structure with exponent ζ_R for relationship-specific inputs and ζ_H for homogeneous inputs;
- recipe-level common productivity draws have mean

$$B_{\omega\rho} b_l^{-\beta_l^\rho} \prod_{\hat{\omega} \in \hat{\rho}} b_{\hat{\omega}}^{-\beta_{\hat{\omega}}^\rho}, \quad (10)$$

with $\beta_l^\rho + \sum_{\hat{\omega} \in \hat{\rho}} \beta_{\hat{\omega}}^\rho = \gamma$.

Define normalized recipe shares

$$\alpha_L^\rho = \frac{\beta_l^\rho}{\gamma}, \quad \alpha_{\hat{\omega}}^\rho = \frac{\beta_{\hat{\omega}}^\rho}{\gamma}, \quad \alpha_R^\rho = \sum_{\hat{\omega} \in \hat{\rho}_R} \alpha_{\hat{\omega}}^\rho, \quad \alpha_H^\rho = \sum_{\hat{\omega} \in \hat{\rho}_H} \alpha_{\hat{\omega}}^\rho. \quad (11)$$

- **Interpretation.** α_R^ρ and α_H^ρ summarize how heavily recipe ρ relies on relationship-specific versus homogeneous materials.
- **Why ζ_R matters.** The elasticity of substitution across potential supplier sets for a given relationship-specific input is $\zeta_R + 1$. A low ζ_R means few close substitutes, so contracting problems are more painful.

3.5 Equilibrium cost index and aggregation

Proposition 1. Industry ω has cost index

$$C_\omega = \left\{ \sum_{\rho \in \mathcal{R}_\omega} \kappa_{\omega\rho} B_{\omega\rho} \left[(t_x^*)^{\alpha_R^\rho} t_l^{\alpha_L^\rho} \prod_{\hat{\omega} \in \hat{\rho}} C_{\hat{\omega}}^{\alpha_{\hat{\omega}}^\rho} \right]^{-\gamma} \right\}^{-1/\gamma}, \quad (12)$$

where

$$t_x^* = \left(\int_1^\infty t_x^{-\zeta_R} dT(t_x) \right)^{-1/\zeta_R}. \quad (13)$$

- **Interpretation.** C_ω is an equilibrium unit-cost index. Court frictions enter through t_x^* , which summarizes both the **direct** cost of distorted customized inputs and the **indirect** cost of being forced toward worse suppliers or worse recipes.
- **Network meaning.** A higher cost of contract-sensitive inputs propagates through all downstream industries that use them directly or indirectly.

Proposition 2. Aggregate consumption is increasing in labor supply and decreasing in the industry cost indices. Up to a constant,

$$U \propto \left(\sum_{\omega \in \Omega} v_\omega J_\omega^{\frac{\eta-1}{\varepsilon-1}} C_\omega^{1-\eta} \right)^{\frac{1}{\eta-1}} L. \quad (14)$$

- **Interpretation.** Aggregate productivity losses are entirely summarized by how bad courts raise the vector of industry cost indices $\{C_\omega\}$.

3.6 The factor-share result that identifies distortions

Proposition 3. Among firms that produce with recipe ρ , average expenditure shares satisfy

$$\text{share on relationship-specific input } \hat{\omega} \in \hat{\rho}_R = \frac{\alpha_{\hat{\omega}}^\rho}{t_x}, \quad (15)$$

$$\text{share on homogeneous input } \hat{\omega} \in \hat{\rho}_H = \alpha_{\hat{\omega}}^\rho, \quad (16)$$

$$\text{share on labor} = \alpha_L^\rho + \left(1 - \frac{1}{\bar{t}_x} \right) \alpha_R^\rho, \quad (17)$$

with

$$\bar{t}_x = \left(\int_1^\infty t_x^{-1} d\tilde{T}(t_x) \right)^{-1}, \quad \tilde{T}(t_x) = \frac{\int_1^{t_x} t^{-\zeta_R} dT(t)}{\int_1^\infty t^{-\zeta_R} dT(t)}. \quad (18)$$

- **Key identification implication.** At the *recipe level*, average factor shares do not depend on relative prices; they depend only on technology parameters and distortions.
- **Why this is useful.** The authors do not need comparable plant-level input prices. They infer wedges from cross-state differences in average cost shares for plants using the same recipe.
- **Economic intuition.** Relationship-specific inputs look artificially cheap in the data because part of their effective cost shows up as extra labor needed to fix customization problems.

3.7 Welfare and counterfactual formulas

If court quality changes so that the wedge distribution shifts from T to T' , then

$$\frac{U'}{U} = \left[\sum_{\omega} HH_{\omega} \left(\frac{C'_{\omega}}{C_{\omega}} \right)^{1-\eta} \right]^{1/(\eta-1)}, \quad (19)$$

where HH_{ω} is the household expenditure share of industry ω . Industry costs satisfy

$$\left(\frac{C'_{\omega}}{C_{\omega}} \right)^{-\gamma} = \sum_{\rho \in \mathcal{R}_{\omega}} R_{\omega\rho} \left[\left(\frac{t_x^{*'}}{t_x^*} \right)^{\alpha_R^{\rho}} \prod_{\hat{\omega} \in \hat{\rho}} \left(\frac{C'_{\hat{\omega}}}{C_{\hat{\omega}}} \right)^{\alpha_{\hat{\omega}}^{\rho}} \right]^{-\gamma}. \quad (20)$$

The first-order version is especially intuitive:

$$d \log U = - \sum_{\omega} HH_{\omega} d \log C_{\omega}, \quad (21)$$

$$d \log C_{\omega} = \sum_{\rho \in \mathcal{R}_{\omega}} R_{\omega\rho} \left[\alpha_R^{\rho} d \log t_x^* + \sum_{\hat{\omega} \in \hat{\rho}} \alpha_{\hat{\omega}}^{\rho} d \log C_{\hat{\omega}} \right]. \quad (22)$$

- **Interpretation.** A better court lowers t_x^* , which lowers upstream costs, which lowers downstream costs, and so on through the production network.
- **Envelope insight.** To first order, the welfare effect depends on exposure weights and the network, not on the higher-order substitution parameters γ and η .

3.8 Structural identification and recipe clustering

Proposition 5. Let s_j^R , s_j^H , and s_j^L be plant j 's spending on relationship-specific inputs, homogeneous inputs, and labor as shares of revenue. Then for firms producing output ω with recipe ρ ,

$$\mathbb{E} \left[\bar{t}_x \frac{s_j^R}{\alpha_R^{\rho}} - \frac{s_j^H}{\alpha_H^{\rho}} \right] = 0, \quad \mathbb{E} \left[\frac{s_j^L + s_j^R}{\alpha_L^{\rho} + \alpha_R^{\rho}} - \frac{s_j^H}{\alpha_H^{\rho}} \right] = 0. \quad (23)$$

The structural estimation imposes:

- **Assumption 3.** Recipe technologies are common across states, while wedges vary by state.
- **Assumption 4.** In state d , wedges are Pareto:

$$T_d(t_x) = 1 - t_x^{-\tau_d}, \quad t_x \geq 1. \quad (24)$$

This implies

$$\bar{t}_{x,d} = 1 + \frac{1}{\zeta_R + \tau_d}, \quad t_{x,d}^* = \left(\frac{\tau_d + \zeta_R}{\tau_d} \right)^{1/\zeta_R}. \quad (25)$$

The recipe-classification step uses Ward's clustering algorithm:

$$\min_{\rho_n \preceq \rho_{n-1}} \sum_{\rho \in \rho_n} \sum_{j \in \rho} \sum_{\hat{\omega}} (m_{j\hat{\omega}} - m_{\rho\hat{\omega}})^2, \quad (26)$$

followed by probabilistic assignment

$$P(j \in \rho) = \frac{[\sum_{\omega'} (m_{j\omega'} - m_{\rho\omega'})^2]^{-1/2}}{\sum_{\rho' \in \rho(\omega)} [\sum_{\omega'} (m_{j\omega'} - m_{\rho'\omega'})^2]^{-1/2}}. \quad (27)$$

- **Interpretation.** Recipes are not imposed ex ante. They are recovered from recurring clusters in plant input mixes, then estimated jointly with the wedges.
- **Algorithm.** Start with a guess for state wedges, cluster plants into recipes after removing those wedges, re-estimate recipe parameters and wedges from the moment conditions, and iterate to convergence.

The remaining parameter ζ_R is estimated from domestic-versus-import sourcing:

$$\log \frac{X_{\omega\hat{\omega}t}^{DOM}}{X_{\omega\hat{\omega}t}^{IMP}} = \zeta \log(1 + \iota_{\hat{\omega}t}) + \lambda_t + \lambda_{\omega\hat{\omega}} + \eta_{\omega\hat{\omega}t}. \quad (28)$$

- **Estimate used in the counterfactual.** $\zeta_R = 0.218$ for the sample of relationship-specific inputs.
- **Implication.** The elasticity of substitution across sets of relationship-specific suppliers is only about 1.218, so finding a close substitute supplier is hard.
- **Calibration.** The paper sets $\eta = 1$ and $\gamma = 1$ and shows the counterfactuals are not very sensitive to them.

4 Identification and instruments

4.1 Baseline identification idea

- **Reduced-form comparison.** Compare plants facing different court congestion across states, especially in industries with higher reliance on relationship-specific inputs.
- **Structural comparison.** Compare average cost shares *within recipes* across states. Since recipe technologies are held constant across states, systematic differences in first moments are interpreted as wedges.
- **Key maintained assumption.** Weak enforcement distorts relationship-specific inputs and labor, but not homogeneous inputs. This is conservative: if homogeneous inputs were also distorted, the true wedges would be larger.

4.2 Instrumental-variable strategy

The core IV uses the age of the High Court (and its interactions with industry relationship specificity) as an instrument for court congestion.

- **Relevance.** Newly created courts start fast and build backlogs over time. The first-stage relationship is visible in Figure II, and the paper reports an F -statistic of about 11.1.
- **Historical logic.** Many newer High Courts were created when new states were split off from old ones, often for political or ethno-nationalist reasons rather than because of current manufacturing composition.

- **Why exclusion is plausible.** The creation of new courts is argued to be orthogonal to contemporaneous plant-level input bundles, once the paper controls for income and state characteristics.
- **Why remaining violations are likely conservative.** Newly formed states tend to be poorer and lower-capacity. That should, if anything, make firms use more homogeneous inputs, which biases the IV estimate toward zero rather than away from it.

4.3 Addressing endogeneity concerns

- **Reverse causality.** The authors argue that firm-to-firm contract disputes are a small share of all cases, so it is unlikely that manufacturing input choices themselves are driving court congestion.
- **Omitted state characteristics.** The regressions interact relationship specificity with log GDP per capita, trust, corruption, caste fragmentation, and language fragmentation.
- **Time variation.** The appendix also uses changes in court pendency and new court benches over time; the results line up with the cross-sectional evidence.
- **Structural heterogeneity concern.** The iterative cluster-estimation procedure is designed precisely to separate technology heterogeneity from state-specific wedges.

5 Tables and figures

5.1 Figure 1: Heterogeneity in input mixes within narrow industries

Read:

- The figure plots plant-level input mixes for single-product plants producing bleached cotton cloth and polished diamonds.
- The key message is that even within a five-digit product, firms use dramatically different input bundles. For polished diamonds, some plants buy cut diamonds and only polish, while others buy rough diamonds and both cut and polish.
- This is the empirical motivation for the recipe concept: a one-production-function-per-industry model would miss economically meaningful organizational heterogeneity.

5.2 Figure 2: Age of the High Court and speed of enforcement

Read:

- Older High Courts have older pending civil cases. Goa and Sikkim are fast; Uttar Pradesh and West Bengal are slow.
- This is the main first-stage picture for the IV strategy.
- The visual is important because it ties modern court congestion to historical court formation rather than to current manufacturing conditions.

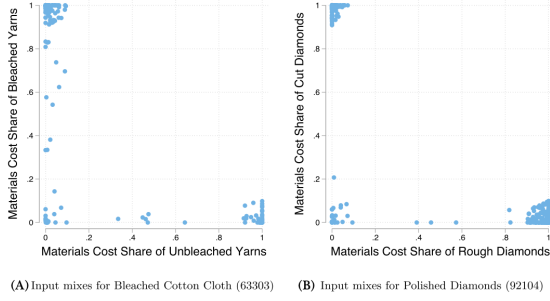


FIGURE I
Heterogeneity in Input Mixes within Narrow Industries

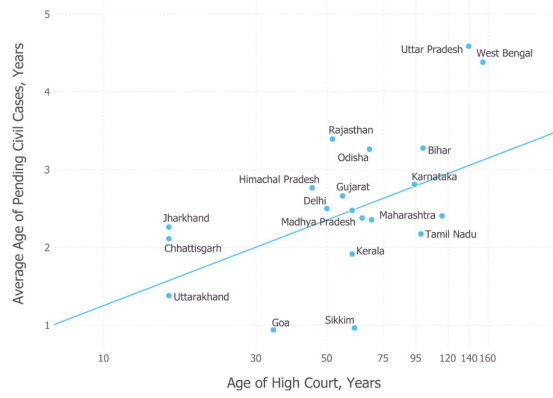


FIGURE II
Age of the High Court and Speed of Enforcement

5.3 Table 1: Materials shares and court congestion (Fact 1)

Read:

- The interaction $\text{CourtCong} \times \text{RelSpec}$ is negative in every specification.
- Baseline OLS is about -0.0167 ; the fully controlled IV estimate is about -0.0212 .
- Interpretation: each additional year of court congestion lowers the materials share by roughly 1.2–2.1 percentage points more in industries that rely on relationship-specific inputs.
- Economic meaning: poor courts push firms away from contract-sensitive outsourcing toward primary inputs and in-house activity.

	Dependent variable: Materials expenditure in total cost					
	(1)	(2)	(3)	(4)	(5)	(6)
Avg age of civil cases * Rel. spec.	-0.0167^{***} (0.0046)	-0.0129^* (0.0051)	-0.0118^* (0.0053)	-0.0156^* (0.0085)	-0.0201^* (0.0082)	-0.0212^{**} (0.0078)
LogGDP/capita * Rel. spec.		0.0114 (0.0086)	0.0102 (0.0091)	0.00710 (0.0095)	0.00556 (0.0096)	
Rel. spec. * State controls			Yes		Yes	Yes
Five-digit industry FE	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	IV	IV	IV
R ²	0.480	0.482	0.484	0.480	0.482	0.484
Observations	208,527	199,544	196,748	208,527	199,544	196,748

Notes: Standard errors are in parentheses, clustered at the state $\times$ industry level. * $p < .10$, ** $p < .05$, *** $p < .01$. Rel. spec. * State controls are interactions of trust, language Herfindahl, caste Herfindahl, and corruption with relationship specificity. The sample consists of single-product plants only.

	Dependent variable: $\frac{x_{ij}^R}{x_{ij}^T + x_{ij}^R}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Avg age of civil HC cases	-0.00547^* (0.0022)	-0.00621^{**} (0.0023)	-0.00530^* (0.0024)	-0.0144^{**} (0.0044)	-0.0146^{**} (0.0044)	-0.0167^{***} (0.0045)
Log district GDP/capita		-0.00389 (0.0045)	-0.00384 (0.0046)		-0.00912^* (0.0051)	-0.00980^* (0.0051)
State controls			Yes			Yes
Five-digit industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	IV	IV	IV
R ²	0.441	0.446	0.449	0.441	0.446	0.449
Observations	225,590	204,031	199,339	225,590	204,031	199,339

Notes: The dependent variable is the share of relationship-specific inputs in total materials cost of plant j . "State controls" are trust, language Herfindahl, caste Herfindahl, and corruption. The sample consists of single-product plants only. Standard errors are in parentheses, clustered at the state $\times$ industry level. * $p < .10$, ** $p < .05$, *** $p < .01$.

5.4 Table 2: Input mix and court congestion (Fact 2)

Read:

- The dependent variable is the share of relationship-specific inputs in total materials cost.
- Coefficients on court congestion are negative throughout, from about -0.005 in OLS to about -0.017 in IV.
- Interpretation: slower courts tilt the materials basket away from customized inputs and toward homogeneous, spot-market inputs.
- This is the cleanest reduced-form evidence that enforcement affects *which* inputs firms buy, not just *how much* they buy.

5.5 Table 3: Vertical span of plants and court congestion (Fact 3)

TABLE III
VERTICAL SPAN OF PLANTS AND COURT CONGESTION (Fact 3)

	Dependent variable: Vertical span					
	(1)	(2)	(3)	(4)	(5)	(6)
Avg age of civil cases * Rel. spec.	0.0195 ⁺ (0.011)	0.0269* (0.012)	0.0280* (0.012)	0.0292 (0.019)	0.0314 ⁺ (0.018)	0.0368* (0.018)
LogGDP * Rel. spec.		0.0464* (0.022)	0.0288 (0.024)		0.0491* (0.023)	0.0330 (0.024)
Rel. spec. * State controls			Yes			Yes
Five-digit industry FE	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	IV	IV	IV
R ²	0.443	0.451	0.453	0.443	0.451	0.453
Observations	163,334	156,191	154,021	163,334	156,191	154,021

Notes: Standard errors are in parentheses, clustered at the state * industry level. * p < .10, ** p < .05, *** p < .01. Rel. spec. * State controls are interactions of trust, language Herfindahl, caste Herfindahl, and corruption with relationship specificity. The sample consists of single-product plants only.

TABLE IV
STATISTICS ON PRODUCTS AND RECIPES

	Count
Products (five-digit ASIC)	4,530
Products with $\geq$ three plants	3,573
Products with $\geq$ five plants	3,034
Recipes	26,776
Recipes with $\geq$ three plants	6,280
Recipes with $\geq$ five plants	2,574
Avg. plants per recipe	8.2
Std. dev. plants per recipe	79.4

Notes: "Products" are the five-digit product codes in our data, "recipes" are the output from our clustering procedure. Plant counts include only single-product plants.

Read:

- The interaction $\text{CourtCong} \times \text{RelSpec}$ is positive across specifications, ranging from about 0.020 to 0.037.
- Interpretation: where courts are slower, plants in relationship-specific industries perform more upstream stages themselves.
- This is the organizational counterpart to the materials-share result: bad courts encourage vertical integration or at least a broader in-house span of production.

5.6 Tables 4 and 5 plus Figure 3: Recipe statistics and classification

TABLE V
SUMMARY STATISTICS ON RECIPES

	Mean	Std. dev.	Min	Max
Cost share of L	0.40	0.22	0.0002	0.999
Cost share of X_R	0.27	0.28	0	0.999
Cost share of X_H	0.33	0.30	0	0.998
Number of inputs with cost share > 1%	4.4	4.6	1	37
Number of inputs with cost share > 0.1%	6.4	12.6	1	205

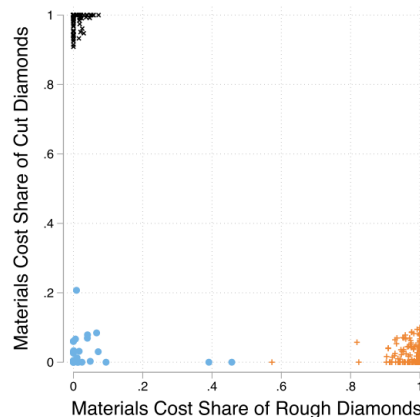


FIGURE III

Recipe Classification Example: Polished Diamonds (92104)

The figure shows the result from the recipe classification procedure (after correcting for wedged) for "polished diamonds" (92104). Observations that are classified as belonging to different recipes are tagged with different markers.

Read:

- Table 4 shows how much heterogeneity the clustering algorithm uncovers: 4,530 five-digit products but 26,776 estimated recipes, with an average of 8.2 plants per recipe.
- Table 5 shows that recipes vary substantially in cost shares: average labor share is 0.40, relationship-specific materials 0.27, and homogeneous materials 0.33.

- Figure 3 shows the recipe classification for polished diamonds. The clusters line up with economically sensible production modes, which strengthens the case that “recipes” are not a statistical artifact.

5.7 Figure 4: Correlation between estimated $\bar{t}_x$ and court congestion

Read:

- This figure takes the structural wedges estimated from the model and plots them against court congestion.
- States with slower courts have larger estimated distortions on relationship-specific inputs. The OLS and IV fitted lines are similar.
- The figure is the bridge from reduced-form facts to the structural counterfactual: the wedges extracted from first moments line up with an independently measured institutional variable.

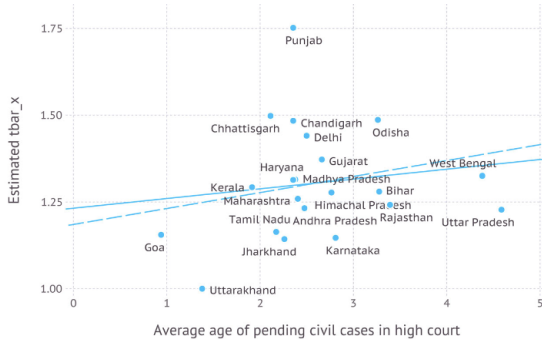


FIGURE IV
Correlation between $\bar{t}_x$ and Court Congestion

The figure shows the correlation between $\bar{t}_x$ and the average age of pending civil cases in the state's high court. The solid line is the OLS regression line; the dashed line is the fit of an IV regression where the age of pending cases is instrumented using the log age of the high court.

TABLE VI
ESTIMATING ζ

	Dependent variable: $\log\left(\frac{x_{\text{DOM}}}{x_{\text{MF}}}\right)$		
	(1)	(2)	(3)
$\log(1 + t_{\text{rel}})$	0.617 (0.44)	0.218 (0.77)	1.209* (0.52)
Industry $\times$ Input FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Level	Five-digit	Five-digit	Five-digit
Sample	All inputs	R only	H only
R^2	0.601	0.580	0.623
Observations	23,692	12,002	11,690

Notes. Robust errors are in parentheses, clustered at the state $\times$ industry level. * $p < .10$, * $p < .05$, ** $p < .01$. Dependent variable is the log ratio of total expenditure on domestically sourced to total imported intermediate inputs of type ω among producers of ω at time t . We only use census plants (which are surveyed every year) to reduce artificial fluctuations that result from sampling.

5.8 Table 6: Estimating ζ

Read:

- The tariff-based regression implies $\zeta_R \approx 0.218$ for relationship-specific inputs.
- That is a low value, implying a low elasticity of substitution across sets of suppliers for customized inputs.
- Quantitatively, this matters because poor courts do not just tax a given supplier match; they also make it hard for firms to replace a problematic supplier with a nearly equivalent one.

5.9 Figure 5: Counterfactual increases in aggregate productivity

Read:

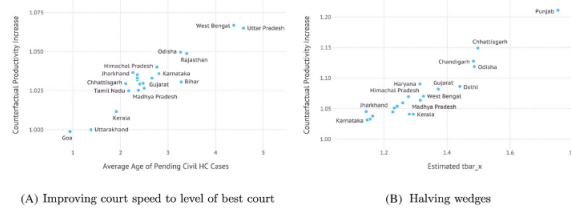


FIGURE V
Counterfactual Increases in Aggregate Productivity

- Panel A reduces wedges by the part statistically explained by court congestion, down to about one year of pending-case age. Productivity gains are positive for essentially all states and average around 4%.
- Panel B halves the wedges directly. Average gains are 7.1%, and the most distorted states see gains well above 10% and in some cases above 20%.
- This is the paper’s macro punchline: the contract-enforcement distortions identified from plant-level input shares are large enough to matter at the aggregate level.

6 Short summary

- **Conclusion.** Weak contract enforcement in India distorts both the level and composition of intermediate-input use and pushes firms toward broader in-house production.
- **Intuition.** When customized supplier contracts are hard to enforce, firms avoid relationship-specific inputs and substitute toward standardized inputs or more vertical integration.
- **Empirical lesson.** To study misallocation in input markets, it is not enough to assume one common production function per industry; one must account for technologically meaningful heterogeneity in recipes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Court congestion materially distorts sourcing and organization decisions in Indian manufacturing.
- **Magnitude.** An extra year of congestion reduces materials shares in contract-sensitive industries, shifts input baskets toward homogeneous inputs, and increases plants’ vertical span.
- **Credibility.** The results survive rich controls, an IV based on the age of High Courts, and a structural estimation strategy that conditions on recipe-level heterogeneity.
- **Mechanism evidence.** The distorted margin is specifically the use of relationship-specific inputs, exactly where formal enforcement should matter most.
- **Quantitative implication.** Better courts would raise aggregate manufacturing productivity by economically meaningful amounts—about 4% on average when congestion is improved to best-state levels, and 7.1% on average when wedges are halved.

- **Additional discipline.** The model separates direct enforcement costs from indirect losses that arise when firms substitute toward worse suppliers or worse technologies. The indirect component is nontrivial but smaller than the direct one.

7.2 Economic intuition

- Contract enforcement is an input-market institution. It determines whether firms can safely buy customized intermediate goods from outside suppliers.
- If courts are weak, outside suppliers become risky. Firms then economize on those inputs, not because the technology no longer wants them, but because the institutional environment makes them expensive in shadow terms.
- Since production is sequential and networked, these distortions do not stop at a single firm. They alter the equilibrium organization of production and raise costs downstream throughout the manufacturing system.

7.3 Takeaway

This paper is a strong example of how to combine reduced-form institutional variation, detailed production data, and structural network modeling. Its big lesson is that misallocation in developing economies can arise not only from taxes or financial frictions, but also from weak enforcement in markets for inputs. Once firms are allowed to choose recipes and supplier networks, court quality becomes a first-order determinant of both plant organization and aggregate productivity.